SVKM'S NMIMS Deemed to be UNIVERSITY
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING™

FOSS MPSTME

HF'26

ADVANCED TRACK · TEAM VISIONX

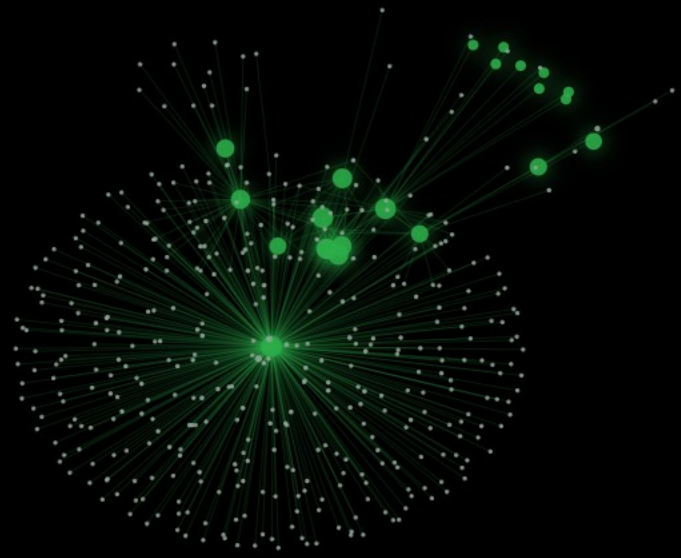
PROBLEM STATEMENT: GHOST ATTACKS

TEAM NAME: VISIONX

IDEA TITLE: UNMASK

A quantitative **OSINT + blockchain engine**. It links anonymous wallets to the person behind them, with evidence a human can check, and finds the stolen money **before it is cashed out**.

LIVE → unmask-five.vercel.app



real trace · Euler Finance hack, March 2023 · 442 addresses

PROBLEM FRAMING

Act inside the cash-out window, without accusing the wrong person.



OSINT investigators

forum posts · usernames · GitHub code

matched by hand in spreadsheets: too slow

Blockchain analysts

wallets · transfers · timestamps

Objective

Wallets → person → money, with a reason for every clue

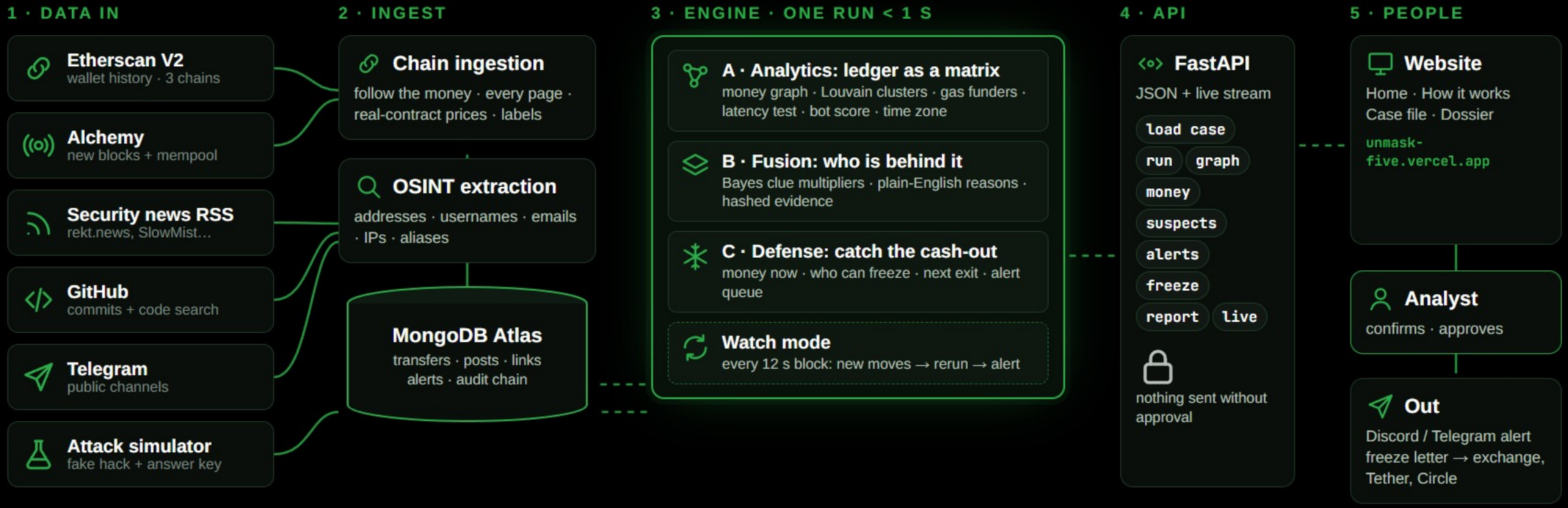
Core trade-off

Speed vs. certainty: act inside the cash-out window

Failure modes we avoid

Wrong accusations · inflated numbers · missed window

ARCHITECTURE & TECH STACK



BACKEND · ENGINE

`Python 3.11` `FastAPI` `NetworkX` `SciPy` `pandas` `scikit-learn`

DATA

`MongoDB Atlas` `Etherscan V2` `Alchemy` `GitHub API` `RSS`

WEBSITE

`React 19` `Vite` `three.js` `GSAP` `Lenis`

DEPLOY · COST

`Render` `Vercel` `200 tests` `$0 · free tiers`

MODEL AND ALGORITHM

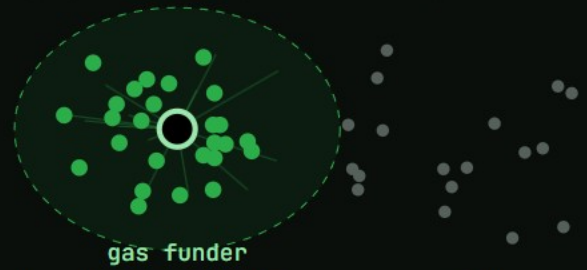
STEP 1
Cluster wallets
money graph + shared gas funders + bridge matching

STEP 2
Latency test
do their posts come just before the moves?

STEP 3
Bayesian fusion
every clue multiplies the odds, with a reason

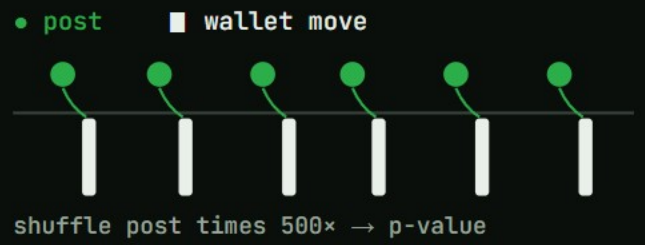
STEP 4
Catch the cash-out
money now · who can freeze · next exchange

WHO OWNS WHICH WALLETS



Fresh wallets funded by the **same payer** belong together

POSTS VS. MOVES



Attack 7: **32 of 33** posts land minutes before a move.
By chance: **~2 in 1,000**

HOW A SUSPECT'S SCORE IS BUILT

1% start ×30 ×30 ×20 ×10 ×3

- knew an attacker address
- named on-chain
- posts precede moves
- posted the exploit code
- same time zone

→ **ghostfox: high confidence**, still "suspected" until a person confirms

- ✓ No labelled hacks to train a model on
- ✓ Every clue can be challenged
- ✓ Confidence halves after a week with no news

HANDLING UNCERTAINTY AND EDGE CASES

Bots mimic market noise

- ✓ **Bot score**: timing test, round amounts, Benford's law

Fake tokens, poisoning

- ✓ Priced by **real contract**: Euler $272M → true $191.6M

Flash loans

- ✓ Only wallets **loaded whole** count as holding money

Researchers quote the address

- ✓ Public addresses **don't count**: 23% → baseline

One small cash-out

- ✓ No prediction until **≥5%** of the money shows it

API errors, rate limits

- ✓ **Retry + backoff**; errors never cached

No public trail at all

- ✓ **Names nobody**: real hacks show no suspect

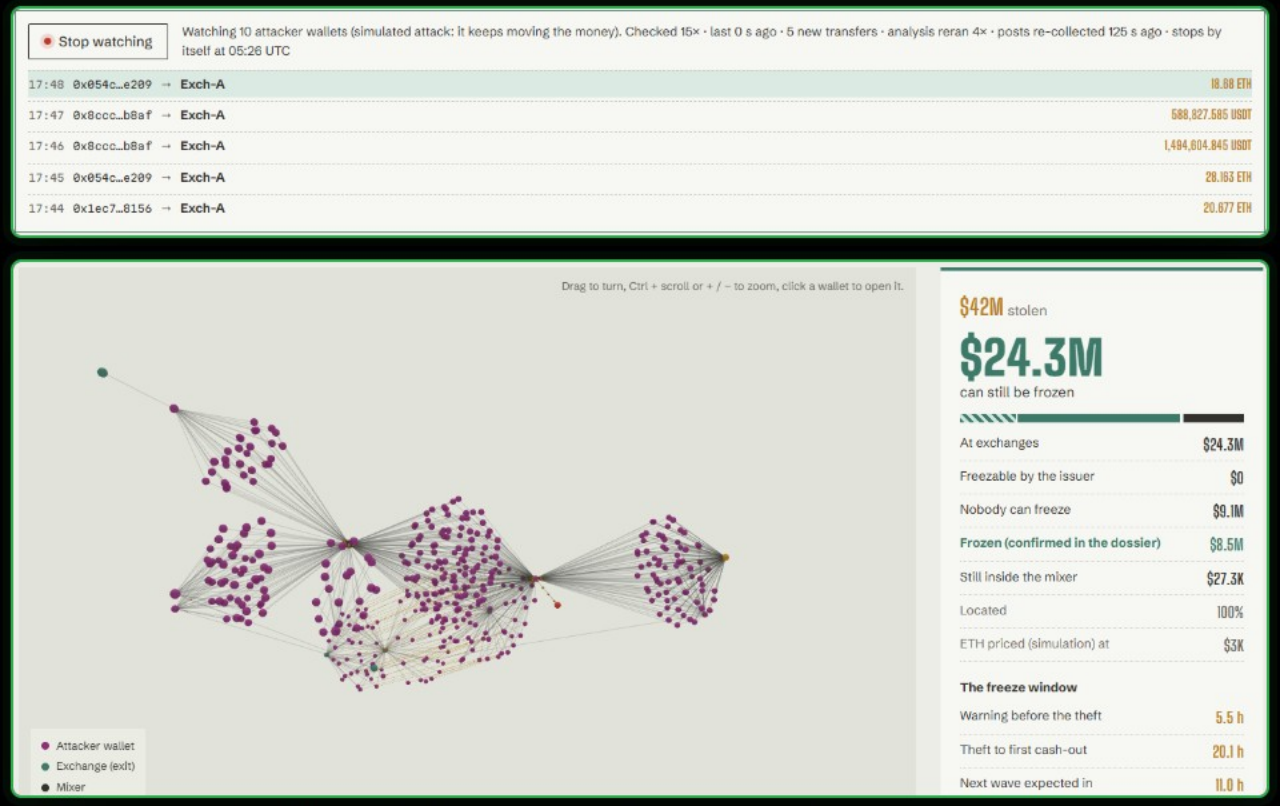
Emails and IPs found

- ✓ **Masked** until a reason is recorded

WHEN UNMASK IS WRONG

Suspected, not accused → **Analyst confirms or rejects** → **Nothing sent without approval** → **Every decision in a tamper-evident audit chain** → **Confidence halves after a week**

LIVE DEMO / SIMULATION RUN



Real screenshots: live feed, money graph, money panel

1. Load attack 7
$42M · 9,055 transfers · 572 posts

2. Analyse in ~1 s
100% precision · ghostfox, 5 clues

3. Watch live
every block: new move → "Just now… freeze it" alert

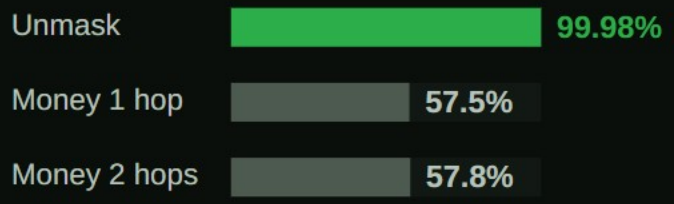
4. Freeze
letter to Tether / exchange → marked frozen

Try it: unmask-five.vercel.app

RESULTS

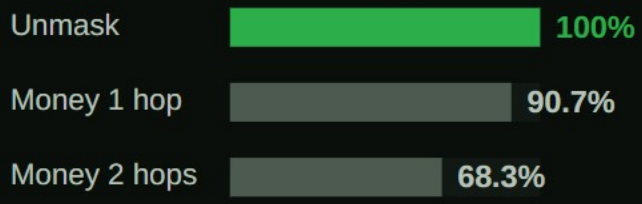
RECALL · WALLETS FOUND

share of the attacker's wallets found, 20 simulated attacks



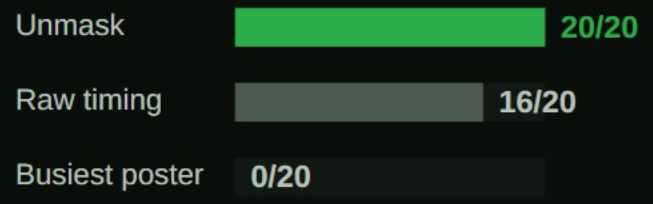
PRECISION · NO FALSE WALLETS

share of flagged wallets that really are the attacker's



RIGHT PERSON · WITH A DECOY

a fake account posting **5× more** than the attacker



REAL HACKS, CHECKED AGAINST THE PUBLIC RECORD

Mar 2023
Euler Finance
$191.6M
traced · $189.4M returned



Jun 2022
Harmony bridge
$98.1M
into Tornado Cash



Apr 2022
Beanstalk
$74.4M
traced



Nov 2023
KyberSwap
$33.5M
traced on 3 chains

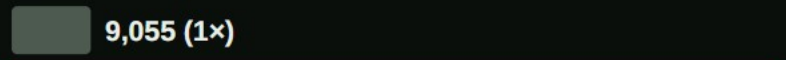


TRADE-OFFS

WE GAINED	WE CHOSE	WE GAVE UP
✓ every score has reasons	**EXPLAINABLE STATISTICS**	⚠ subtle patterns a trained model might catch
✓ honest, verifiable numbers	**WHOLE-WALLET BALANCES**	⚠ undercounts in huge pools (KyberSwap)
✓ instant loads on a fresh server	**RECORDED REAL TRACES**	⚠ those 4 cases are a snapshot
✓ $0 to run	**FREE TIER**	⚠ ~25 s wake-up · slower on the server
✓ no automatic accusations	**HUMAN SENDS LETTERS**	⚠ a few minutes of speed

UNDER 10× LOAD (MEASURED)

Transfers analysed



Analysis time on a laptop



`accuracy unchanged: 100% precision, right person`

Next: update per block instead of re-running everything

LIMITS AND NEXT STEPS

Today
KNOWN WEAKNESSES

- Identity proven on simulated data
- Unpriced tokens (WBTC) not counted
- No login: one shared case
- Watch stops when the server sleeps

+1 week
WHAT WE'D ADD

- Login, one case per analyst
- Always-on watch worker
- Prices for more tokens
- Labelled real OSINT cases

Production
WHAT IT NEEDS

- Roles + audited access
- Own node / indexer + job queue
- Incremental analysis per block
- Legal process with exchanges